Nirupama Patra

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SUMMARY

Software Engineer with 5+ years of commercial experience building production-quality, user-facing features using React, Next.js, TypeScript, and JavaScript (ES6+), with hands-on experience in state management (Redux, Zustand, Context/Hooks) and unit testing (Jest). Comfortable owning features end-to-end, from Figma to deployment, collaborating closely with Product Owners and backend engineers using Git, code reviews, and CI/CD (GitHub Actions, Azure Pipelines) in Agile teams. Working experience across back end (Node.js, Python, REST/GraphQL APIs) and data engineering (SQL-based validation, schema design), with production systems deployed and monitored on Azure.

PROFESSIONAL EXPERIENCE

Software Engineer, VE3, Maidenhead, UK ***Sep '25 – Present***

- Built responsive frontend features in TypeScript and Next.js, using Zustand for state management, translating Figma designs into the GOV.UK Design System, enabling consumer and B2B users to search, apply advanced filters, and upload/download bulk data.
- Structured the frontend as an npm/Yarn workspaces monorepo (apps/*, packages/*), sharing components, utilities, and configuration across multiple front-end applications.
- Developed and maintained backend REST and GraphQL APIs as part of a distributed, microservices-based architecture in Node.js, securing sensitive data with OAuth2/JWE authentication and designing Azure SQL/PostgreSQL schemas, allowing internal teams and third-party developers to reliably submit and retrieve large volumes of data, including a delta-paginated API and an asynchronous bulk-export feature.
- Deployed and maintained CI/CD pipelines using Azure Pipelines and GitHub Actions, enabling frequent, reliable releases across both platforms with minimal manual intervention.
- Strengthened platform reliability by writing Jest unit tests, setting up Azure Application Insights monitoring, and supporting production incident resolution (Zoho Desk) within Agile sprints, reducing the time to catch and fix issues before they reached users.

Frontend Developer & AI Automation Expert (Intern), Re:Healthify, London, UK ***Jun '25 – Aug '25***

- Architected a backend automation platform integrating LangChain and LLM agents to extract structured data from unstructured banking and financial documents, reducing manual data entry time by 40% and minimizing manual review as new records were onboarded.
- Built a data pipeline that scrapes wellness provider information from online sources, stores it in PostgreSQL, and serves it via a REST API, and implemented the corresponding React frontend to power a directory web app, enabling users to discover nearby gyms, yoga studios, and wellness centers by category, with map-based navigation, user reviews, and location switching in development.

Application Developer (Banking & Finance), IBM India Private Ltd. ***Oct '19 – Sep '23***

- Built and maintained frontend components with React, TypeScript, and Redux for state management, delivering intuitive, client-facing dashboards that gave customers real-time, secure access to their accounts and transactions, reducing reliance on manual support for account queries.
- Ingested and processed client-supplied data files into a DB2 database, using SQL to filter, validate, and append records against a data dictionary to meet client-specific formatting and quality requirements, and built reusable, object-oriented data-access and validation modules so this validated data could reliably power the secure backend REST APIs consumed by those dashboards.
- Maintained platform reliability through automated testing and CI/CD pipelines, while contributing to sprint planning, code reviews, and mentoring junior engineers in a global Agile Scrum team.

EDUCATION

MSc in Advanced Computer Science, Cardiff University — Distinction ***Sep '23 – Dec '24***

B.Tech in Computer Science and Engineering, Haldia Institute of Technology — 8.58/10 ***Aug '15 – Jun '19***

PERSONAL PROJECTS

Bengali Lexical Simplification (Dissertation Project)

- Built an NLP-based lexical simplification system for Bengali text using BERT, RoBERTa, and mBERT, with a complex word identification pipeline (word frequency, syllable count, POS tagging, semantic features) and a helper module for candidate refinement; evaluated across multiple benchmark datasets. Tech: Python, Hugging Face, Transformers, NLP.

Encode London – Quant Track - Finalist: Developed a full-stack crypto trading strategy platform utilizing TypeScript and web frameworks, integrating automated technical indicator engines and real-time data pipelines.

Mistral AI Hack London – Participant: Built a medical AI assistant application leveraging PIXTRAL-12B and a custom RAG pipeline with LangChain for vector data retrieval and context-aware clinical responses.

KEY SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), Next.js, HTML5, CSS3

State Management: Zustand, Redux, React Hooks & Context API

Backend & APIs: Node.js, Express.js, Fastify, Python, REST & GraphQL API design, OAuth2/JWE authentication, object-oriented design

AI-Assisted Engineering: LangChain, RAG, LLM agents, AI-assisted development tooling

Quality & Delivery: Unit testing (Jest), code review, CI/CD (GitHub Actions, Azure Pipelines), monorepo tooling (npm/Yarn workspaces), Git, Agile/Scrum, production support

Databases: PostgreSQL, Azure SQL, SQL Server, DB2, schema design

CERTIFICATIONS

Databricks Certified Data Engineer Associate · Microsoft AZ-900 (Azure Fundamentals) · IBM Mainframe Technologies · SQL (Udemy) · COBOL & SAS (Udemy) · Java & C (Oracle & Aptech) · Python (Coursera – University of Michigan) · Big Data (IBM Developer Skills Network)